

Online Resource 2: Reproduction Manifest

Version: replication package v1.0.0, 2026-08-31

This manifest separates scholarly identification of research objects from the exact identifiers needed to reproduce the reported analysis. The manuscript cites the SWE-bench paper, Verified construction account, and versioned software repositories. The full identifiers below pin the exact materials.

Frozen upstream sources

Resource	Scholarly/software citation	Exact version	Retrieval location
SWE-bench Verified canonical task identifiers	Jimenez et al. (2024); OpenAI (2024)	500-task princeton-nlp/SWE-bench_Verified test split	URLs enumerated in source_manifest.json
Outcome matrices	SWE-bench Team (2026a)	experiments commit 1faa91cade0562ba62b66c1c99e71f7b72d96f13	https://github.com/swe-bench/experiments/tree/1faa91cade0562ba62b66c1c99e71f7b72d96f13
Leaderboard scores and metadata	SWE-bench Team (2026b)	website commit f42505b21a0eb31a9cc1204caafcb0da6c1a259	https://github.com/swe-bench/swe-bench.github.io/tree/f42505b21a0eb31a9cc1204caafcb0da6c1a259

Collection timestamp recorded by the authoritative run: 2026-08-26T13:14:23.499910+00:00.

Source matrix digests

Matrix	SHA-256
Open/classic binary outcome matrix	e477915c5dd68a132995f692da67b0105743f34bd868f636d3b5fec43c1b11e0
Standardized Bash-only binary outcome matrix	0f5fda63360d75604589e4916057ffc294dd3b4ef6d9cca9552adf651806357d

The binary matrices are not redistributed. source_manifest.json lists every immutable API/raw URL required to recollect them and records redistributes_upstream_outcome_matrix: false.

Authoritative experiment execution

Field	Value
Git repository	https://github.com/coloursophia/coding-agent-benchmark-maintenance
Tagged release	v1.0.0
Persistent DOI	https://doi.org/10.5281/zenodo.22189000
Experiment commit	4bcbd4a2cd259f9722e1fa3eb83fa1e03b79df75
GitHub Actions run	32970788181
Workflow URL	https://github.com/coloursophia/coding-agent-benchmark-maintenance/actions/runs/32970788181
Completion date	2026-08-26
Workflow duration	23 min 38 s
Downloaded artifact SHA-256	c66da9edd849c36335fb15a687331d2058cb8c900c29264d38b2d06b2070c334
Formal configuration	configs/formal.json
Python recorded by source manifest	3.12.14

Required result files

The authoritative artifact contains the following files; the companion verification script checks that every listed relative path exists:

- formal_metrics.csv (208 panel-scope-method-budget rows);
- harmonized_metrics.csv (208 corresponding harmonized rows);

- secondary_metrics_online_resource.csv (208 complete secondary-metric rows);
- harmonized_decisions.csv, budget_stability.csv, and threshold_sensitivity.csv;
- fixed_selection_decisions.csv, fixed_selection_sensitivity.csv, and selection_overlap.csv;
- cluster_mapping.csv, cluster_sensitivity.csv, and cluster_sensitivity_decisions.csv;
- curve_band_diagnostics.csv, curve_bootstrap_stability.csv, bootstrap_stability.csv, and random_curve_coupling_sensitivity.csv;
- longitudinal.csv, formal_results.json, source_manifest.json, data_quality.json, analysis_history.md, summary.md, and report.html.

Online Resource 1 is Online_Resource_1_secondary_metrics.csv, SHA-256 1f7603e22c4469a39a19459ea63f6c4593301da50bb5c178ae9a7dc979854b26.

Verification commands

From the repository root:

```
python -m unittest discover -s tests -v
python manuscript/scripts/sync_result_tables.py --artifact artifacts/github-run-32970788181/unpacked --check
python manuscript/scripts/verify_online_resource_2.py --artifact artifacts/github-run-32970788181/unpacked
```

The release workflow additionally renders the final DOCX to PDF/PNG, inspects every page, audits accessibility and styles, and confirms that all author-year citations and reference entries are bidirectionally matched. The three commands above were executed verbatim from a clean repository snapshot; the saved transcript is Online_Resource_2_clean_verification_v3.3.txt.

Persistent archive

The public version 1.0.0 release is archived by Zenodo under the persistent DOI <https://doi.org/10.5281/zenodo.22189000>. The archive carries the MIT license, the four-author citation metadata, the source and reproduction manifests, and the generated aggregate outputs. Exact upstream commits and checksums remain listed above so that the scholarly citation and byte-level reproduction chain serve distinct purposes.